

Zero-Shot Generalization in Visual RL

MIPT Internship Test Assignment

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Prologue

In vision-based Reinforcement Learning, agents often struggle to generalize outside the training distribution. Most approaches rely on heavy data augmentation or massive datasets. In this project, you will explore an alternative: the *Associative Latent DisentAnglement* (ALDA) algorithm, which uses disentangled representations combined with associative memory to enable zero-shot out-of-distribution generalization without augmentation.

1 INTRODUCTION

Generalization in vision-based RL remains a central challenge. Conventional approaches such as DrQ [7], SVEA [3], or SAC+AE [8] depend on augmentations that approximate unseen variations by modifying training images. While effective, such techniques require enormous compute and data. ALDA [1] proposes a different route: it learns a *disentangled latent representation* (via QLAE [4]) and equips it with an *associative memory mechanism* to remap out-of-distribution factors back to in-distribution values. In doing so, ALDA demonstrates zero-shot adaptation on challenging benchmarks such as the Distracting Control Suite.

This test assignment will immerse you in reproducing and analyzing ALDA.

2 THE ASSIGNMENT

1. Carefully read the paper [1]: *Zero-Shot Generalization of Vision-Based RL Without Data Augmentation*.
2. Implement the ALDA algorithm on top of Soft Actor-Critic (SAC) [2], faithfully following the paper's description and pseudocode.¹
3. Train your ALDA agent on selected tasks from the **DeepMind Control Suite** [6] and evaluate it on OOD variants from the **Distracting Control Suite** [5].
4. Demonstrate whether the agent exhibits any *zero-shot generalization ability* to distribution shifts.
5. Prepare a report in **ICML style**,² including:
 - a clear description of the algorithm and your implementation,

¹We recommend starting from the SAC implementation in CleanRL: <https://github.com/vwxyzjn/cleanrl>, though you are free to choose another implementation source.

²ICML L^AT_EX template available at: <https://www.overleaf.com/latex/templates/icml2025-template/dhxrkcgvnkt>

- training and evaluation results (plots, learning curves, visualizations of latent traversals if possible),
 - an analysis of OOD generalization performance,
 - and a link to your code repository.
6. In your report, pose one simple research question you encountered (e.g., “How sensitive is ALDA to latent dimensionality?”) and attempt to answer it based on your experiments.
 7. If you cannot complete all experiments due to time or compute limits, submit what you have. Clearly describe what you tried, what worked, and what remains open.

Evaluation Criteria

Your submission will be evaluated on:

- **Implementation correctness:** clean, documented, faithful implementation of ALDA.
- **Experimental effort:** training curves, evaluation in at least one Distracting Control Suite setting.
- **Analysis:** clarity and depth in interpreting whether zero-shot generalization emerges.
- **Presentation quality:** professional ICML-style report with effective visualizations.

Do not worry if your numbers fall short of the paper — the focus is on clarity, reproducibility, and insight.

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